EMILY R. BURSK

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Education

University of Cincinnati — Cincinnati, OH
Bachelor of Science in Astrophysics; GPA: 3.5 — *Aug. 2023 – May 2025*

Cincinnati State Technical and Community College — Cincinnati, OH
Associate of Science; GPA: 3.4 — *Aug. 2020 – May 2023*

Research Experience

Space Science Institute — Boulder, CO
Postbaccalaureate Researcher — *Mar. 2026 - Present*

- Working with Dr. Ben Horowitz to reconstruct large-scale structure from cosmological simulations using differentiable modeling techniques.

Thomas More University — Crestview Hills, KY
Undergraduate Researcher — *July 2024 - July 2025*

- Worked with Dr. Austin Hinkel to develop and implement analyses using the Illustris-TNG50 simulation to study how the relationship between stellar structure and dark matter distribution influences galactic warping.
- Presented research findings at the 245th American Astronomical Society Conference.

University of Cincinnati — Cincinnati, OH
Undergraduate Researcher — *June 2023 - Aug. 2024*

- Worked with Dr. Matthew Bayliss to develop Python scripts for the JWST NIRSpec IFU science calibration pipeline to enhance spectral data reduction efficiency and processed strongly lensed galaxy observations into calibrated data cubes.
- Led undergraduate researchers on JWST data reduction methods for additional lensed targets.

Cincinnati State Technical and Community College — Cincinnati, OH
Undergraduate Researcher — *Jan. 2022 - Jan. 2023*

- Composed a review paper on strong gravitational lensing and its applications in astrophysical research.
- Presented findings at the Ohio Aerospace Institute student research symposium.

Employment and Experience

University of Colorado Boulder — Boulder, CO
Student Grader, ASTR 1200: Stars and Galaxies — *Sep. 2025 - Dec. 2025*

- Evaluated homework assignments and exams for an introductory astronomy course with 105 students.
- Provided feedback to support students' understanding of key astronomy concepts.

National Aeronautics and Space Administration (NASA) — Cleveland, OH
Student Observer, Here to Observe (H2O) Program — *Sep. 2023 - May 2024*

- Attended science mission meetings for the Europa Clipper mission.
- Attended the 55th Lunar and Planetary Science Conference.
- Collaborated with mentor on a project comparing atmospheric models of Europa, contributing to the understanding of the moon's environment.

Cincinnati Observatory Center — Cincinnati, OH
Program Educator — *Jan. 2024 - June 2025*

- Delivered 25+ astronomy presentations annually, adapting content from introductory to advanced levels for audiences ranging from children to adult learners.

- Led STEM outreach programs at schools and community events, expanding access to science education.

Education Intern *Aug. 2022 - Dec. 2023*

- Operated the oldest public telescope in North America for public viewings.
- Supported outreach programs for underserved schools, providing hands-on STEM learning experiences.

Publications

Refereed Publications:

1. Rigby, J. R., Vieira, J. D., Phadke, K. A., Hutchison, T. A., Welch, B., Cathey, J., Spilker, J. S., Gonzalez, A. H., Adhikari, P., Aravena, M., Bayliss, M. B., Birkin, J. E., **Bursk, E.**, et al. (2024). *JWST Early Release Science Program TEMPLATES: Targeting Extremely Magnified Panchromatic Lensed Arcs and Their Extended Star Formation.* arXiv:2312.10465

Conference Proceedings:

3. **Bursk, E.**, Hinkel, A. (5 April 2025). Linking Dark and Stellar Substructure in the Illustris TNG50 Simulation [Poster presentation]. Kentucky Area Astronomical Society Meeting, Berea, KY, USA.
2. **Bursk, E.**, Hinkel, A. (15 January 2025). Linking Dark and Stellar Substructure in the Illustris TNG50 Simulation [Poster presentation]. American Astronomical Society Meeting 245, National Harbor, MD, USA.
1. **Bursk, E.**, Huber, J. (31 March 2023). Discovering the Distant Universe Using Strong Gravitational Lensing [Poster presentation]. 2023 Student Research Symposium, Cleveland, OH, USA.

Scientific Communication:

4. **Bursk, E.** (2025). "Let's Chat!: Emmy Bursk". Girls in STEM Magazine.
3. **Bursk, E.**, Hinkel, A. (2025). "Linking Dark and Stellar Substructure in the Illustris TNG50 Simulation", Senior Capstone Report, University of Cincinnati
2. **Bursk, E.** (2024). "Failure to Launch: Emily Bursk" [Podcast episode]. Future-Proof Podcast.
1. Hewald, R., Younker, N., Adhikari, P., Elicker, L., **Bursk, E.** (2023). "A Getting Started Guide for JWST NIRSpec IFU Data Reduction"

Public Talks

5. **Bursk, E.** (2025). "Nature's Telescopes: Gravitational Lensing". In: Cincinnati Observatory Astronomy Class
4. **Bursk, E.** (2025). "The Moon: Origins and Exploration". In: Cincinnati Observatory Nearest Neighbors Night
3. **Bursk, E.** (2024). "Europa Clipper: NASA's Mission to Jupiter's Ocean Moon". In: Cincinnati Observatory Jupiter Night
2. **Bursk, E.** (2023). "Icy Moons: The Search for Life". In: Cincinnati Observatory Astronomy Evenings
1. **Bursk, E.** (2023). "Tour of the Universe". In: Cincinnati Observatory Astronomy Evenings

Programming Skills

Languages: Python 3.x (Astropy, NumPy, Pandas, SciPy)

Software: Jupyter, LaTeX, Mathematica, SAOImageDS9

Honors and Awards

UC Cincinnati Pathways Scholarship	2023-2025
University of Cincinnati Dean's List	2023-2025
Cincinnati State Dean's List	2021-2023
Ohio Space Grant Consortium Community College Scholarship	2022

Volunteer Experience

Denver Astronomical Society — Denver, CO
Member/Volunteer — *Jan. 2026 - Present*

- Develop and deliver educational talks at community astronomy events on astronomical concepts and current research topics.
- Assist with public astronomy nights and open house events, guiding guests in telescope observations and explaining observed celestial objects.

Cincinnati Astronomical Society — Cincinnati, OH
Member/Volunteer — *Sep. 2022 - June 2025*

- Contributed to a radio astronomy project through Stanford University's Space Weather Monitor program.
- Certified on operating a 14-inch telescope, using it to locate and observe deep sky objects, stars, and planets.